

Extended sign-kill for $\widehat{\lambda} = (4, 2)$: the E_j^- chain at $j \leq q - 3$ via three-branch support

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Abstract

Let $\lambda = (4, 2, 1^q) \vdash n = q+6$ with $q \geq 3$, $\ell = q+2$, and $B = B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} R'_{\ell+3} R'_{\ell+4} V_\lambda$. We prove the “ $\subseteq$ ” direction of the extended sign-kill conjecture for this family at $q \geq 4$:

$$B \subseteq \bigcap_{j=1}^{q-3} E_j^- |_{V_\lambda},$$

matching the predicted threshold $\tau(\lambda) = q+3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q-3$. We further prove sharpness at the boundary $j = q-2$ (structural reduction + computational witness for $q \in \{4, 5, 6, 7\}$). At $q = 3$ the containment is vacuous (the predicted intersection is empty).

The proof follows the recipe established in the previous (very late prove session, fifth pass) paper on $\widehat{\lambda} = (3, 2)$. Three new ingredients are needed:

1. An extension of the May-20 three-branch reduction to $\lambda = (4, 2, 1^q)$, which involves *four* R' -factors in Ω and hence *three* outer (T_j+1) factors after Φ_X -extraction (rather than two as in $\widehat{\lambda} = (3, 2)$).
2. Application of the May-14 fifth-pass extended sign-kill theorem for $\widehat{\lambda} = (3, 2)$ *recursively* to the inner piece $\mu_A = (3, 2, 1^{q-1})$, which requires $q \geq 4$.
3. Application of the hook support lemma (May-14 fifth pass, Theorem 3) to the inner piece $\mu_B = (4, 1^q)$, which requires $q \geq 4$ ($m \geq k$ at $k = 4$, $m = q$).

The μ_C -piece vanishes by the same column-1 leakage mechanism as in the May-20 paper. Combined with the universal $B \subseteq E_\ell^+$ (May-13 evening), this closes the $\widehat{\lambda} = (4, 2)$ row of the May-14 evening conjecture table in the containment direction at $q \geq 4$.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n-1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q) |_{V_\lambda} = \text{Im}(S_j |_{V_\lambda}), \quad E_j^- := \ker(S_j) |_{V_\lambda}.$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad B := \Omega \cdot V_\lambda = B_{\ell+1}^{(\lambda)} V_\lambda.$$

For $\lambda = (4, 2, 1^q)$, $n = q+6$ and $\ell = q+2$, so $\Omega = R'_{q+3} R'_{q+4} R'_{q+5} R'_{q+6}$ has *four* R' -blocks (equivalently, $\Omega = R'_{n-3} R'_{n-2} R'_{n-1} R'_n$). By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda(\geq 2)} = f^{(3,1)} = 3$ in the stable regime $q \geq 2$.

Seminormal support. For $W \subseteq V_\lambda$, define

$$\text{supp}(W) := \{T \in \text{SYT}(\lambda) : \text{some } w \in W \text{ has } [v_T]w \neq 0\}.$$

Column-1 identity prefix. For an SYT T of any partition μ , the *identity prefix length* is

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

2 Input from prior papers

2.1 The support rule and prefix corollary (May-14 fourth pass)

Proposition 1 (Support rule). *Let $W \subseteq V_\lambda$ be any subspace. Fix $1 \leq j \leq n-1$. If every $T \in \text{supp}(W)$ has j and $j+1$ in the same column of T , then $W \subseteq E_j^-|_{V_\lambda}$.*

Remark 2. If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P-1$ the entries $j, j+1$ sit at column-1 rows $j, j+1$, hence share column 1. By Proposition 1, $W \subseteq E_j^-|_{V_\lambda}$ for $1 \leq j \leq P-1$.

2.2 The May-14 fifth-pass results

We need both of the May-14 fifth-pass theorems.

Theorem 3 (Extended sign-kill for $\hat{\lambda} = (3, 2)$, May-14 fifth pass). *For $\lambda' = (3, 2, 1^{q'}) \vdash n' = q' + 5$ with $q' \geq 3$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 6 = q' - 1$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 2$.*

This is Theorem 6 (Main) of 2026-05-14-extended-sign-kill-32.tex together with the prefix-bound construction in its proof of Step 1–Step 3.

Theorem 4 (Hook support lemma, May-14 fifth pass). *Let $\mu = (k, 1^m) \vdash n_0 := k + m$ with $k \geq 2$ and $m \geq k$. Set $\ell_0 := \ell(\mu) = m + 1$ and $j_0 := \ell_0 + 1 = m + 2$. Then every $T \in \text{supp}(B_{j_0}^{(\mu)} V_\mu)$ satisfies $p(T) \geq m - k + 2$.*

This is Theorem 3 of 2026-05-14-extended-sign-kill-32.tex.

2.3 The outer-factor preservation lemma (May-14 fifth pass)

Lemma 5 (Outer-factor preservation). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P+1$, every $T'' \in \text{supp}((T_j+1)W)$ also satisfies $p(T'') \geq P$.*

This is Lemma 4 of 2026-05-14-extended-sign-kill-32.tex. The proof is elementary: (T_j+1) acts on the seminormal basis within $\{v_T, v_{s_j T}\}$; the swap s_j moves only entries j and $j+1$, both $\geq j \geq P+1 > P$, so neither lies in the column-1 prefix and the prefix is preserved.

2.4 The May-13 hook column-1 result (rank-zero)

We will use the May-12 hook rank-zero theorem:

Theorem 6 (Hook column-1, May-12). *For $\mu = (k, 1^m) \vdash n_0$ with $k \geq 2$ and $m \geq \max(k, 2)$, $B_{\ell_0+1}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$.*

This is the May-12 hook paper (2026-05-12-hook-j-formula.tex), also restated in the May-20 paper as Theorem 2.3.

2.5 Phase A endpoint (May-19)

Theorem 7 (Abstract Phase A endpoint). *For any H_q -module V and any $1 \leq j \leq n-1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.*

3 The three-branch reduction for $(4, 2, 1^q)$

For $\lambda = (4, 2, 1^q)$ with $q \geq 3$ (so $n \geq 9$, $\ell = q + 2 = n - 4$), the three removable corners are

$$c_1 = (1, 4), \quad c_2 = (2, 2), \quad c_3 = (n-4, 1).$$

The three doubly-stripped partitions are

$$\begin{aligned} \mu_A &:= \lambda \setminus \{c_1, c_3\} = (3, 2, 1^{q-1}), \\ \mu_B &:= \lambda \setminus \{c_2, c_3\} = (4, 1^q), \\ \mu_C &:= \lambda \setminus \{c_1, c_2\} = (3, 1^{q+1}), \end{aligned}$$

each of size $n-2$.

The branching maps Φ_X . Exactly as in the May-20 paper, for each $X \in \{A, B, C\}$ and each $T' \in \text{SYT}(\mu_X)$, define two SYTs T_A^X, T_B^X of λ by extending T' with $\{n-1, n\}$ at the corresponding corner pair in the two possible orders, and set

$$\Phi_X(v_{T'}) := v_{T_A^X} + \tau_X v_{T_B^X},$$

where $\tau_X = \tau(d_X) \in \mathbb{Q}(q)^\times$ is the Hoefsmit T_{n-1} -block coefficient at axial distance $d_X := c(c_X^+) - c(c_X^-)$. The axial distances are

$$|d_A| = |(1-(n-4))-(4-1)| = n-6, \quad |d_B| = |(1-(n-4))-(2-2)| = n-5, \quad |d_C| = |0-3| = 3.$$

All $\geq 3 \geq 2$ for $q \geq 3$, so the T_{n-1} 2-blocks are non-degenerate.

Lemma 8 (Φ_X properties). *For each $X \in \{A, B, C\}$:*

- (i) $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ is injective and T_i -equivariant for $1 \leq i \leq n-3$.
- (ii) The seminormal supports of $\Phi_A(V_{\mu_A}), \Phi_B(V_{\mu_B}), \Phi_C(V_{\mu_C})$ are pairwise disjoint and their direct sum equals $E_{n-1}^+|_{V_\lambda}$.

Proof. The argument is verbatim May-20 Lemmas 3.2–3.4: injectivity from distinct corner-pair extensions; equivariance from the T_{n-1} -commutation with $\{T_1, \dots, T_{n-3}\}$ and the decoupling argument on 4-dimensional joint blocks; disjointness from distinct corner-pair supports; dimension match by counting $+q$ -eigenvectors over the three T_{n-1} -corner-pair classes. The argument depends only on λ having three corners and on the axial distances being ≥ 2 , both of which hold for $\lambda = (4, 2, 1^q)$ at every $q \geq 3$. □ □

Theorem 9 (Three-branch reduction for $(4, 2, 1^q)$). *For $\lambda = (4, 2, 1^q)$ with $q \geq 3$,*

$$B = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \sum_{X \in \{A, B, C\}} \Phi_X(B_{n-4}^{(\mu_X)} V_{\mu_X}). \quad (1)$$

Proof. We follow the May-15/May-20 pull-through argument, adapted to $\Omega = R'_{n-3}R'_{n-2}R'_{n-1}R'_n$ (four R' -factors).

Step 1 (Phase A). By Theorem 7 at $V = V_\lambda$, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$. By Lemma 8(ii), this equals $\bigoplus_X \Phi_X(V_{\mu_X})$.

Step 2 (apply R'_{n-1}). Write $R'_{n-1} = (T_{n-2}+1)U$ where $U := (T_{n-3}+1)\cdots(T_1+1) \in H_q(S_{n-2})$. Since $T_1, \dots, T_{n-3}$ commute with T_{n-1} (index gaps ≥ 2), U preserves $E_{n-1}^+|_{V_\lambda}$. By Lemma 8(i), $U \Phi_X(v) = \Phi_X(R'^{(\mu_X)}_{n-2} v)$ for $v \in V_{\mu_X}$ (here $R'^{(\mu_X)}_{n-2}$ is the analogue of R'_{n-2} inside $H_q(S_{n-2})$, using generators $T_1, \dots, T_{n-3}$). Hence

$$R'_{n-1} R'_n V_\lambda = (T_{n-2}+1) \sum_X \Phi_X(R'^{(\mu_X)}_{n-2} V_{\mu_X}).$$

Step 3 (apply R'_{n-2}). Write $R'_{n-2} = (T_{n-3}+1)S$ where $S := (T_{n-4}+1)\cdots(T_1+1) \in H_q(S_{n-3})$. S commutes with T_{n-2} (index gap ≥ 2), so $R'_{n-2}(T_{n-2}+1) = (T_{n-3}+1)(T_{n-2}+1)S$. And $S \in H_q(S_{n-3}) \subset H_q(S_{n-2})$ acts on $\Phi_X(v)$ as $\Phi_X(R'^{(\mu_X)}_{n-3} v)$. Therefore

$$R'_{n-2} R'_{n-1} R'_n V_\lambda = (T_{n-3}+1)(T_{n-2}+1) \sum_X \Phi_X(R'^{(\mu_X)}_{n-3} R'^{(\mu_X)}_{n-2} V_{\mu_X}).$$

Step 4 (apply R'_{n-3}). Write $R'_{n-3} = (T_{n-4}+1)S'$ where $S' := (T_{n-5}+1)\cdots(T_1+1) \in H_q(S_{n-4})$. S' commutes with both T_{n-3} (gap = 2) and T_{n-2} (gap = 3), so $R'_{n-3}(T_{n-3}+1)(T_{n-2}+1) = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)S'$. $S' \in H_q(S_{n-4}) \subset H_q(S_{n-2})$ acts on $\Phi_X(v)$ as $\Phi_X(R'^{(\mu_X)}_{n-4} v)$. Combining:

$$B = \Omega V_\lambda = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \sum_X \Phi_X(R'^{(\mu_X)}_{n-4} R'^{(\mu_X)}_{n-3} R'^{(\mu_X)}_{n-2} V_{\mu_X}).$$

The triple product inside is exactly $B^{(\mu_X)}_{n-4} V_{\mu_X}$ (three consecutive R' -factors at indices $n-4, n-3, n-2$ in V_{μ_X} , which has $n-2$ letters), and (1) follows. $\square$ $\square$

4 The C -piece vanishes

Lemma 10 (C -piece vanishing). $B^{(\mu_C)}_{n-4} V_{\mu_C} = 0$ for $\lambda = (4, 2, 1^q)$ with $q \geq 3$.

Proof. $\mu_C = (3, 1^{q+1})$ has length $\ell(\mu_C) = q+2 = n-4$ and size $n-2$, so $j_0(\mu_C) = \ell(\mu_C)+1 = n-3$. By Theorem 6 (applied at $k=3$, $m=q+1 \geq 4$, so $m \geq k=3$), $B^{(\mu_C)}_{n-3} V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$.

Now

$$B^{(\mu_C)}_{n-4} V_{\mu_C} = R'^{(\mu_C)}_{n-4} B^{(\mu_C)}_{n-3} V_{\mu_C}.$$

The rightmost factor of $R'^{(\mu_C)}_{n-4}$ is (T_1+1) , which annihilates $E_1^-|_{V_{\mu_C}}$. Hence $B^{(\mu_C)}_{n-4} V_{\mu_C} = 0$. $\square$ $\square$

Remark 11. As in the May-20 paper for $(3, 2, 1^q)$, the C -corner pair $(\{c_1, c_2\})$ avoids the column-1 bottom corner c_3 . This strips two cells from λ without consuming the column-1 bottom, producing a hook μ_C that is one row *shorter* than μ_A, μ_B (i.e., $\ell(\mu_C) = q+2$ vs. $\ell(\mu_A) = \ell(\mu_B) = q+1$). The shorter μ_C has its j -formula at $j_0(\mu_C) = n-3$ rather than $n-4$; the index $n-4$ at which we evaluate sits one step *below*, and the leading (T_1+1) of $R'^{(\mu_C)}_{n-4}$ kills the E_1^- -contained result. This is the same structural mechanism as in $\hat{\lambda} = (3, 2)$.

By Lemma 10, the reduction (1) simplifies for $\lambda = (4, 2, 1^q)$, $q \geq 3$, to

$$B = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \left(\Phi_A(B^{(\mu_A)}_{n-4} V_{\mu_A}) + \Phi_B(B^{(\mu_B)}_{n-4} V_{\mu_B}) \right). \quad (2)$$

5 Main theorem

Theorem 12 (Main). For $\lambda = (4, 2, 1^q) \vdash n = q + 6$ with $q \geq 4$,

$$B \subseteq \bigcap_{j=1}^{q-3} E_j^-|_{V_\lambda}.$$

Proof. By Remark 2, it suffices to show that every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 2 = n - 8$. We use the reduction (2) together with prefix bounds on the two inner pieces and Lemma 5 on the three outer factors.

Step 1: prefix bound on the Φ_A -image. $\mu_A = (3, 2, 1^{q-1})$ has size $n - 2 = q + 4$, so $n' = q + 4$ and $q' = q - 1$. Since $q \geq 4$ we have $q' \geq 3$, so Theorem 3 applies and gives

$$p(T') \geq n' - 6 = q - 2 = n - 8 \quad \text{for every } T' \in \text{supp}(B_{n-4}^{(\mu_A)} V_{\mu_A}).$$

(Here $B_{n-4}^{(\mu_A)} = B_{\ell(\mu_A)+1}^{(\mu_A)}$ since $\ell(\mu_A) = q + 1 = n - 5$.)

Now $\Phi_A(v_{T'})$ is supported on two SYTs T_A^A, T_B^A of λ obtained from T' by placing $\{n - 1, n\}$ at $\{c_1, c_3\} = \{(1, 4), (n - 4, 1)\}$ in the two possible orders. Cells of μ_A 's column 1 are at rows $1, 2, \dots, \ell(\mu_A) = n - 5$. The added cell $(1, 4)$ is in row 1 but column 4 (not column 1), and $(n - 4, 1)$ is at row $n - 4 > n - 5$, i.e., below μ_A 's column-1 cells. Hence rows $1, \dots, n - 5$ of column 1 in T_A^A, T_B^A are unchanged from T' , so $p(T_A^A), p(T_B^A) \geq n - 8$.

Step 2: prefix bound on the Φ_B -image. $\mu_B = (4, 1^q)$, a hook with $k = 4$, $m = q$. Since $q \geq 4$ we have $m \geq k$, so Theorem 4 applies and gives

$$p(T') \geq m - k + 2 = q - 2 = n - 8 \quad \text{for every } T' \in \text{supp}(B_{n-4}^{(\mu_B)} V_{\mu_B}).$$

(Again $B_{n-4}^{(\mu_B)} = B_{\ell(\mu_B)+1}^{(\mu_B)}$ since $\ell(\mu_B) = q + 1 = n - 5$.)

Now $\Phi_B(v_{T'})$ is supported on two SYTs T_A^B, T_B^B obtained from T' by placing $\{n - 1, n\}$ at $\{c_2, c_3\} = \{(2, 2), (n - 4, 1)\}$. The added cells are at column 2 and row $n - 4 > n - 5 = \ell(\mu_B)$, so they leave the column-1 prefix of T' intact. Hence $p(T_A^B), p(T_B^B) \geq n - 8$.

Step 3: outer factors preserve the prefix. The three outer factors in (2) are (T_j+1) for $j \in \{n - 4, n - 3, n - 2\}$. Each index satisfies $j \geq n - 4 = (n - 8) + 4 > (n - 8) + 1$, so the hypothesis $j \geq P + 1$ of Lemma 5 (with $P = n - 8$) is met. Applying the lemma three times in sequence, the prefix bound $\geq n - 8$ is preserved.

Conclusion. Combining Steps 1–3 and (2): every $T \in \text{supp}(B)$ has $p(T) \geq n - 8 = q - 2$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq q - 3$. $\square$ $\square$

Remark 13 ($q = 3$). At $q = 3$, the predicted threshold is $\tau = q - 3 = 0$ and the intersection $\bigcap_{j=1}^0 E_j^-$ is the whole module V_λ , so the containment is vacuous. Indeed Theorem 12's hypothesis $q \geq 4$ is exactly the joint requirement of the recursive $(3, 2)$ input ($q' \geq 3$, i.e., $q \geq 4$) and the hook lemma ($m \geq k$, i.e., $q \geq 4$). At $q = 3$, neither inner ingredient applies in its stated form, but the prefix bound $p \geq n - 8 = 1$ holds trivially (every SYT has prefix ≥ 1), so the proof framework collapses to a vacuous statement, consistent with the predicted $\tau = 0$.

6 Sharpness

Lemma 14 (Sharpness reduction). *Suppose $B \subseteq E_{q-2}^-|_{V_\lambda}$, where $\lambda = (4, 2, 1^q)$ with $q \geq 4$. Then every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 1 = n - 7$.*

Proof. Set $j_0 := q - 2 = n - 8$. Suppose for contradiction $T \in \text{supp}(B)$ has $p(T) = n - 8$ exactly. Since $T \in \text{supp}(B)$, there exists $w \in B$ with $C_T := [v_T]w \neq 0$; by hypothesis, $T_{j_0}w = -w$.

Locating entry $n - 7$ in T . By $p(T) \geq n - 8$, the cells $\{(1, 1), \dots, (n - 8, 1)\}$ are filled with entries $\{1, \dots, n - 8\}$. Since $p(T) = n - 8$ exactly, $T(n - 7, 1) \neq n - 7$. The remaining 8 cells of λ are

$$\{(1, 2), (1, 3), (1, 4), (2, 2), (n - 7, 1), (n - 6, 1), (n - 5, 1), (n - 4, 1)\},$$

holding the 8 entries $\{n - 7, n - 6, \dots, n\}$.

We claim entry $n - 7$ is forced to sit at $(1, 2)$:

- Not at $(n - 6, 1), (n - 5, 1), (n - 4, 1)$: column-1 increase forces $T(n - 7, 1) < n - 7$, but no eligible entry $< n - 7$ remains for $(n - 7, 1)$.
- Not at $(1, 3)$: $T(1, 2) < T(1, 3) = n - 7$, so $T(1, 2) \in \{n - 7\} \cap (\text{remaining})$ — only candidate is $T(1, 2) < n - 7$, but all remaining entries are $\geq n - 7$. Impossible.
- Not at $(1, 4)$: same argument, $T(1, 2), T(1, 3) < n - 7$ forced, both impossible.
- Not at $(2, 2)$: column-2 increase forces $T(1, 2) < n - 7$, impossible since $T(1, 2) \geq n - 7$ over the remaining entries.

Hence $T(1, 2) = n - 7$.

Off-diagonal swap. The cells of $j_0 = n - 8$ (at $(n - 8, 1)$) and $j_0 + 1 = n - 7$ (at $(1, 2)$) are in different rows ($n - 8 \neq 1$ for $n \geq 10$, i.e., $q \geq 4$) and different columns. Axial distance:

$$|d| = |c(1, 2) - c(n - 8, 1)| = |1 - (1 - (n - 8))| = n - 8 \geq 2 \quad \text{for } q \geq 4.$$

Hoefsmit's 2-block at distance $|d| \geq 2$ is non-degenerate: $T_{j_0}v_T = \alpha v_T + \beta v_{T^*}$ with $\beta \neq 0$, where $T^* := s_{j_0}T$ swaps entries $n - 8$ and $n - 7$ between cells $(n - 8, 1)$ and $(1, 2)$.

T^* is a valid SYT. After the swap, $(n - 8, 1)$ holds $n - 7$ and $(1, 2)$ holds $n - 8$. Column-1 rows $1, \dots, n - 9$ unchanged $(1, \dots, n - 9)$; $(n - 8, 1) = n - 7 > T(n - 9, 1) = n - 9 \checkmark$; $T(n - 7, 1) > T(n - 8, 1)|_{T^*} = n - 7 \checkmark$ (we have $T(n - 7, 1) \in \{n - 6, \dots, n\}$ from the forced argument above). Row 1: $T(1, 1) = 1 < T(1, 2)|_{T^*} = n - 8 \checkmark$; $T(1, 2)|_{T^*} = n - 8 < T(1, 3) \in \{n - 6, \dots, n\} \checkmark$. Column 2: $T(1, 2)|_{T^*} = n - 8 < T(2, 2) \in \{n - 6, \dots, n\} \checkmark$. So T^* is a valid SYT of λ .

Prefix drop forces $T^* \notin \text{supp}(B)$. $T^*(n - 8, 1) = n - 7 \neq n - 8$, so $p(T^*) \leq n - 9 < n - 8$. By Theorem 12, every $U \in \text{supp}(B)$ has $p(U) \geq n - 8$, hence $T^* \notin \text{supp}(B) \supseteq \text{supp}(w)$.

Coefficient contradiction. We compute

$$[v_{T^*}](T_{j_0}w) = \sum_{U \in \text{supp}(w)} C_U \cdot [v_{T^*}](T_{j_0}v_U).$$

The diagonal contribution $[v_{T^*}](T_{j_0}v_{T^*}) = (\text{an } \alpha\text{-multiple of } C_{T^*})$ requires $T^* \in \text{supp}(w)$ (no). The off-diagonal contribution at v_U requires the T_{j_0} 2-block partner of U to be T^* , i.e., $s_{j_0}U = T^*$, i.e., $U = s_{j_0}T^* = T$. Hence

$$[v_{T^*}](T_{j_0}w) = C_T \cdot \beta \neq 0,$$

while $[v_{T^*}](-w) = -C_{T^*} = 0$. This contradicts $T_{j_0}w = -w$. We conclude no $T \in \text{supp}(B)$ has $p(T) = n - 8$ exactly, hence every such T has $p(T) \geq n - 7$. $\square$ $\square$

Theorem 15 (Sharpness, computationally verified). *For $\lambda = (4, 2, 1^q)$ at every $q \in \{4, 5, 6, 7\}$, $\text{supp}(B)$ contains an SYT T with $p(T) = n - 8 = q - 2$ exactly; consequently $B \not\subseteq E_{q-2}^-|_{V_\lambda}$.*

Proof. Existence of $T \in \text{supp}(B)$ with $p(T) = q - 2$ is verified computationally for $q \in \{4, 5, 6, 7\}$: `find_witness.py` (Section 8) reports exactly 42 such witnesses in each case. An explicit witness at $q = 4$ ($n = 10$, prefix = 2):

$$T(1, \cdot) = (1, 3, 4, 5), \quad T(2, \cdot) = (2, 6), \quad T(3, 1) = 7, \quad T(4, 1) = 8, \quad T(5, 1) = 9, \quad T(6, 1) = 10.$$

Column 1 entries are $(1, 2, 7, 8, 9, 10)$, so $p(T) = 2$ exactly. (Analogous witnesses for $q = 5, 6, 7$ are listed by `find_witness.py`.) Combining with Lemma 14, the assumption $B \subseteq E_{q-2}^-$ would force every $U \in \text{supp}(B)$ to have $p(U) \geq q - 1$, contradicting the witness T with $p(T) = q - 2$. Hence $B \not\subseteq E_{q-2}^-$. $\square$

Remark 16 (Stronger empirical statement). Computational verification (mod $P = 100\,003$, $Q = 11$) at $q \in \{4, 5, 6, 7\}$ shows that $S_{j_0}|_B$ has full rank $3 = \dim B$ at $j_0 = q - 2$, i.e., $B \cap E_{j_0}^- = 0$ (not merely $B \not\subseteq E_{j_0}^-$). Theorem 15 proves the weaker non-containment.

7 Eigenspace summary

Theorem 17 (Eigenspace summary, $\widehat{\lambda} = (4, 2)$). *For $\lambda = (4, 2, 1^q)$ with $q \geq 4$,*

$$B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-3} E_j^-|_{V_\lambda}, \quad B \not\subseteq E_{q-2}^-|_{V_\lambda}.$$

The E_ℓ^+ containment is sharp with scalar action: $T_\ell|_B = q \cdot \text{Id}_B$ (May-13 evening Theorem A).

Proof. E_j^- chain at $j \leq q - 3$: Theorem 12. Non-containment at $j = q - 2$: Theorem 15. E_ℓ^+ containment with scalar action: May-13 evening (universal across all λ in stable regime). $\square$ $\square$

The eigenspace constraints alone do not pin down B : computationally the intersection on the right has dimension strictly $> \dim B = 3$ in every tested case. The dimension formula $\dim B = f^{\lambda^{(\geq 2)}} = f^{(3,1)} = 3$ (May-13 multi-corner) is a sharper fact.

8 Computational verification

Theorem 18 (Verification). *Theorem 17 was verified computationally at $q \in \{2, 3, 4, 5, 6, 7\}$ (i.e., $n \in \{8, 9, 10, 11, 12, 13\}$), mod $P = 100\,003$ at $Q = 11$. In every case:*

- $\dim B = 3$, matching the May-13 multi-corner formula.
- $|\text{supp}(B)|$ stabilizes at 123 for $\lambda = (4, 2, 1^q)$ at $q \geq 4$ (boundary value $|\text{supp}(B)| = 115$ at $q = 3$, 69 at $q = 2$). Cf. $|\text{supp}(B)| = 26$ for $(3, 2, 1^q)$ in the May-14 fifth-pass paper.
- Every $T \in \text{supp}(B)$ has $p(T) \geq q - 2 = n - 8$, with the bound achieved at 42 SYTs per q (the sharpness witnesses, for $q \in \{4, 5, 6, 7\}$).
- $T_j b = -b$ for every $b \in B$ and $1 \leq j \leq q - 3$.
- $S_{q-2}|_B$ has rank $\dim B = 3$ (stronger than Theorem 15; cf. Remark 16).
- The C -piece $B_{n-4}^{(\mu_C)} V_{\mu_C} = 0$ verified at every $q \geq 3$.

- *Inner pieces:* $\dim B^{(\mu_A)} = 2$, $\dim B^{(\mu_B)} = 1$ ($\text{sum} = 3 = \dim B$, *matching dim-additivity*), with $\text{min prefix} = q - 2$ for both.

Proof. Scripts in `~/projects/scratch/2026-05-14-extended-sign-kill-42/`. The script `explore_42_family.py` constructs Hoefsmit seminormal matrices for λ and each μ_X , builds Ω and $\Omega^{(\mu_X)}$, computes supports by row-echelon reduction, and checks S_j -rank by direct linear algebra. The script `find_witness.py` enumerates SYTs in $\text{supp}(B)$ with $p(T) = q - 2$ exactly and prints an explicit witness for each q . □ □

9 Discussion

9.1 What is new

This paper closes the $\widehat{\lambda} = (4, 2)$ row of the extended sign-kill conjecture (May-14 evening) in the containment direction at the predicted threshold $\tau = q - 3$, and proves sharpness at the boundary $j = q - 2$ (modulo the computational input Theorem 15, verified for $q \in \{4, 5, 6, 7\}$). Prior to this paper:

- May-14 fourth-pass: $\widehat{\lambda} = (2, 2)$, all j .
- May-14 fifth-pass: $\widehat{\lambda} = (3, 2)$, all j (containment + sharpness).

9.2 What composes here that didn't before

This is the first case where the recursive ingredient (the May-14 fifth-pass $(3, 2)$ theorem) is invoked as an *input*, applied to the inner μ_A piece of a strictly larger shape. This validates the compositional strategy implicit in the “ r -additivity meta-theorem” (May-13 evening): the $(4, 2)$ proof inherits its prefix bound from $\mu_A = (3, 2, 1^{q-1})$ (today) and $\mu_B = (4, 1^q)$ (hook lemma), each of which inherits from its own ingredients. The May-14 fifth-pass paper closed the $(3, 2)$ row knowing $(2, 2)$ as input; today's paper closes the $(4, 2)$ row knowing $(3, 2)$ as input. This is a clean inductive ladder.

9.3 The hypothesis $q \geq 4$

Both inner ingredients require $q \geq 4$:

- Recursive $(3, 2)$ at $\mu_A = (3, 2, 1^{q-1})$ requires $q' = q - 1 \geq 3$, i.e., $q \geq 4$.
- Hook lemma at $\mu_B = (4, 1^q)$ requires $m \geq k$, i.e., $q \geq 4$.

For $q = 3$ the predicted threshold $\tau = 0$ makes the containment vacuous; for $q = 2$ ($\tau = -1$) likewise. We verified $\dim B = 3$ at $q \in \{2, 3\}$ as well, confirming the stable regime starts no later than $q = 2$. ($q = 1$ is non-stable, $\dim B = 6 = 2 \cdot 3$ — the factor-of-2 row-repeat phenomenon also investigated and refuted as a uniform conjecture in the May-14 morning paper.)

9.4 Toward general $\widehat{\lambda}$

The natural next family is $\widehat{\lambda} = (3, 3)$, i.e., $\lambda = (3, 3, 1^q)$. Here $\widehat{\lambda}_1 = \widehat{\lambda}_2 = 3$, so the predicted threshold is $\tau = q + 3 - 3 - 3 = q - 3$, same as $\widehat{\lambda} = (4, 2)$. The corner structure differs:

- For $\lambda = (3, 3, 1^q)$, $\lambda_1 = \lambda_2 = 3$, so $(1, 3)$ is *not* a removable corner (row 1 = row 2). The removable corners are $c_1 = (2, 3)$ (the bottom of row 2) and $c_2 = (q+2, 1)$ (column-1 bottom). Only *two* corners, so this is an $r = 1$ shape, with $\dim B = f^{\lambda^{(\geq 2)}} = f^{(2, 2)} = 2$.

A two-corner shape has a simpler reduction: $r = 1$ in the multi-corner classification, requiring only the May-15 *single* branching reduction (one Φ , one inner piece). The prediction $\tau = q - 3$ for $\hat{\lambda} = (3, 3)$ should follow from the single-branch version of today's framework, with inner piece $\mu = \lambda \setminus \{c_1, c_2\} = (3, 2, 1^{q-1})$ (recursive!) or possibly $(2, 2, 1^{q-1})$ depending on the corner labels. This is structurally simpler than today's three-branch case and should follow from a single invocation of the $(3, 2)$ recursive theorem.

The next interesting $r = 2$ family is $\hat{\lambda} = (4, 3)$, i.e., $\lambda = (4, 3, 1^q)$. Predicted $\tau = q - 4$. Three corners again ($c_1 = (1, 4)$, $c_2 = (2, 3)$, $c_3 = (q + 2, 1)$), three doubly-stripped:

- $\mu_A = (3, 3, 1^{q-1})$: a $\hat{\lambda} = (3, 3)$ shape (recursive — requires the $\hat{\lambda} = (3, 3)$ result above).
- $\mu_B = (4, 2, 1^{q-1})$: a $\hat{\lambda} = (4, 2)$ shape (recursive — today's result at $q - 1$).
- $\mu_C = (3, 2, 1^q)$: a $\hat{\lambda} = (3, 2)$ shape (recursive — May-14 fifth pass).

None of these is a hook! The hook lemma is not needed for $(4, 3)$; all three branches reduce to non-hook inner pieces, each handled by the recursive extended sign-kill chain. This is the first case where the proof framework is *fully recursive* (no hook input).

9.5 What the iso programme this is not

This paper continues the structural sharpening of B via its T_j -eigenspace containments — a programme orthogonal to (and unaffected by) the iso programme refutations of May-13 and May-14 (morning/afternoon). We learn more about *where* B sits inside V_λ via these eigenspace conditions, without claiming B equals any standard representation-theoretic object.

10 Status

Established.

- Theorem 9: three-branch reduction for $\lambda = (4, 2, 1^q)$, $q \geq 3$.
- Lemma 10: μ_C -piece vanishes.
- Theorem 12: $B \subseteq E_j^-$ for $j \leq q - 3$ at $q \geq 4$.
- Theorem 15: sharpness at $j = q - 2$ for $q \in \{4, 5, 6, 7\}$ (and Lemma 14's structural reduction at all $q \geq 4$).
- Theorem 17: complete eigenspace summary.

Open.

- Structural (non-computational) sharpness across all $q \geq 4$ (same obstacle as the May-14 fifth-pass paper; the residual is existence of a boundary-prefix witness in $\text{supp}(B)$).
- Sharpness at $j \in \{q - 1, q, \dots, n - 1\} \setminus \{\ell\}$ (consistent computationally; argument should be analogous to the May-14 fifth-pass paper).
- The next family $\hat{\lambda} = (3, 3)$ at $\lambda = (3, 3, 1^q)$ (single-branch reduction, recursive on $\mu = (3, 2, 1^{q-1})$).
- $\hat{\lambda} = (4, 3)$ at $\lambda = (4, 3, 1^q)$ (fully recursive three-branch, no hook input).
- General $\hat{\lambda}$, $\ell(\hat{\lambda}) \geq 2$: the support prefix bound $p(T) \geq q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$ across $\text{supp}(B)$ (May-14 fifth-pass Conjecture 9).

Push status. Pending PAT refresh; prior unpushed-commit count on `clio-vega/proofs`: 66 (as of May-14 fifth-pass session).

Scripts.

- `~/projects/scratch/2026-05-14-extended-sign-kill-42/explore_42_family.py` — main verification.
- `~/projects/scratch/2026-05-14-extended-sign-kill-42/find_witness.py` — sharpness witnesses.